

TASK 5

Final Report, Automation & Presentation

Sales Performance Analysis using Sample Superstore Data

ApexPlanet Software Pvt. Ltd. | Final Submission

Executive Summary

This project analyzes 8,827 cleaned sales records covering 03 Jan 2014 to 30 Dec 2017. The analysis produced total sales of **\$819,718.06**, total profit of **\$98,850.43**, 4,725 unique orders and 790 unique customers. The overall profit margin is **12.06%**. The results identify the strongest regions, segments and categories, while highlighting low-profit sub-categories that need management attention.

Dashboard Reference

The submitted Power BI dashboard is represented by the provided dashboard screenshot. The final report below uses the same business-analysis themes and validates the KPIs directly from the supplied CSV.

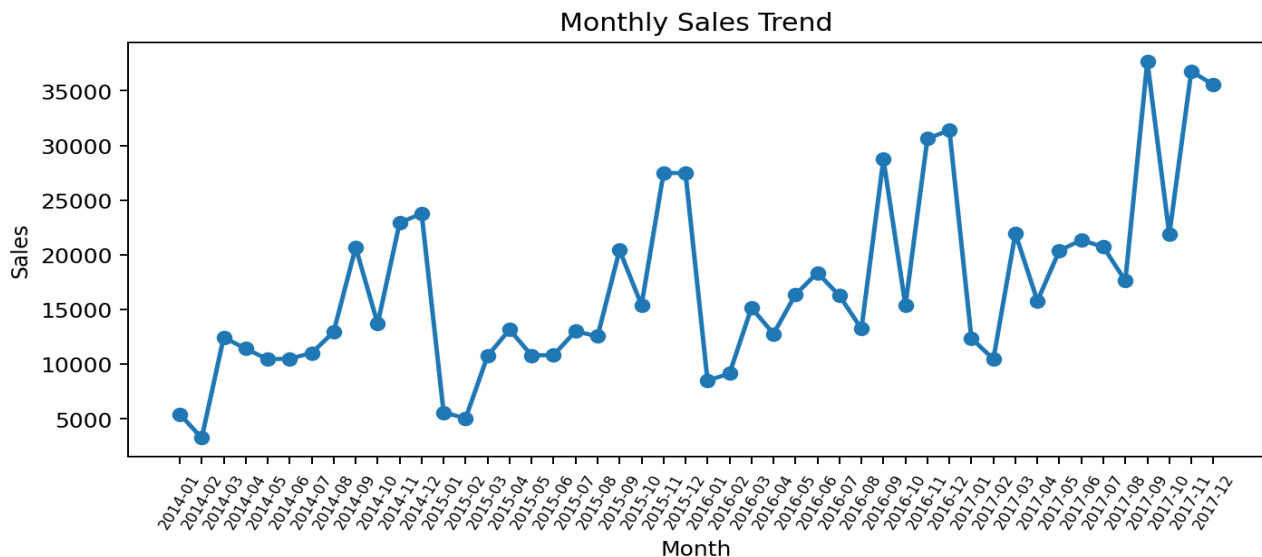
KPI	Result
Total Sales	\$819,718.06
Total Profit	\$98,850.43
Total Orders	4,725
Total Customers	790
Total Quantity	31,853
Average Order Value	\$173.49
Profit Margin	12.06%

1. Data Preparation & Automation

The pipeline removes duplicate rows, standardizes column names, converts dates and numeric columns, handles missing values in selected descriptive fields, and removes records without essential order/date/sales/profit values. Derived fields include Order Year and Order Month.

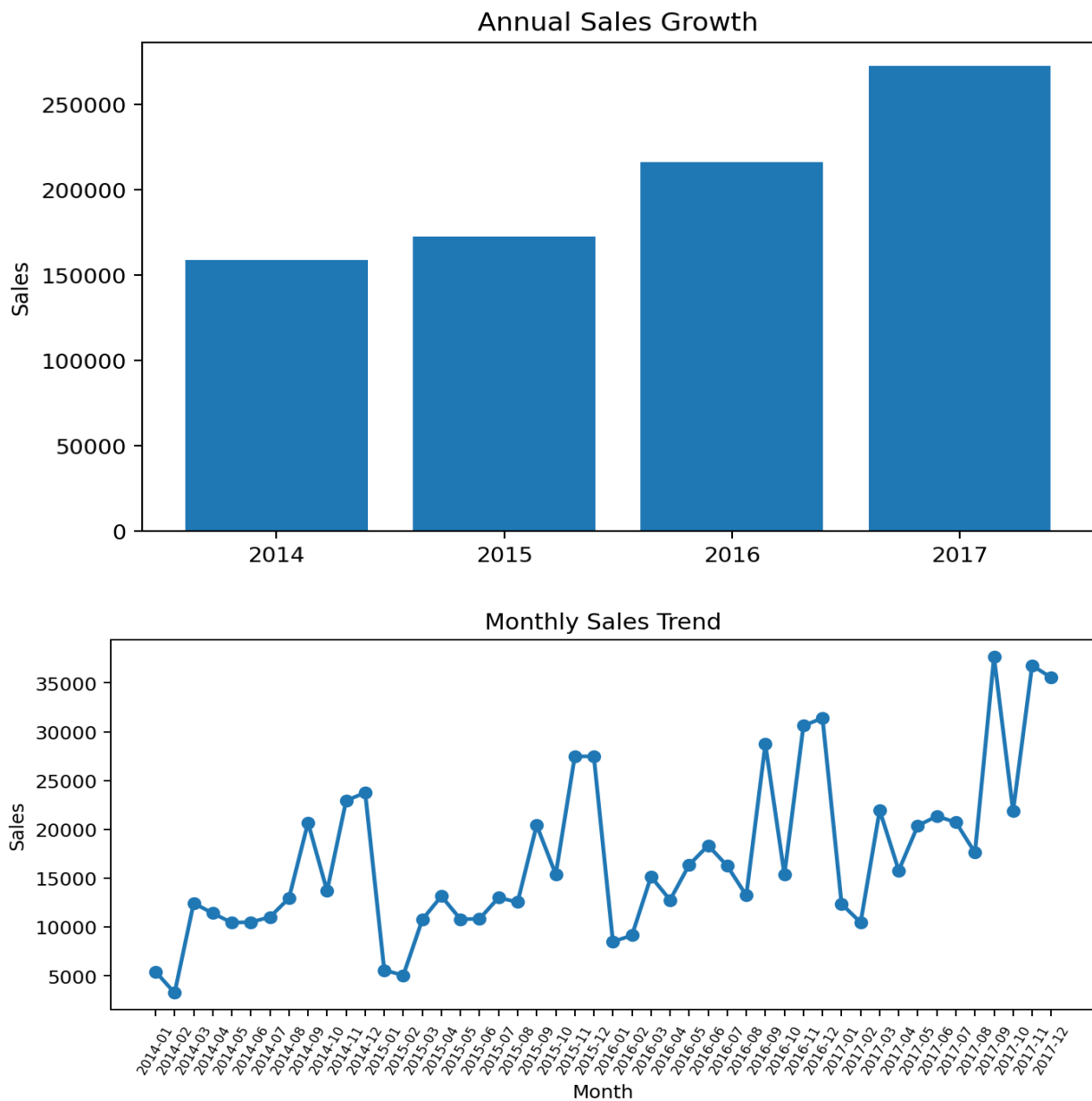
Automation workflow

- Load raw CSV data
- Clean and validate data types
- Remove duplicates and incomplete essential records
- Calculate business KPIs
- Generate category, segment, product and time summaries
- Export processed data and analysis tables to Excel
- Generate charts and KPI summary text
- Run automatically through GitHub Actions on a daily schedule

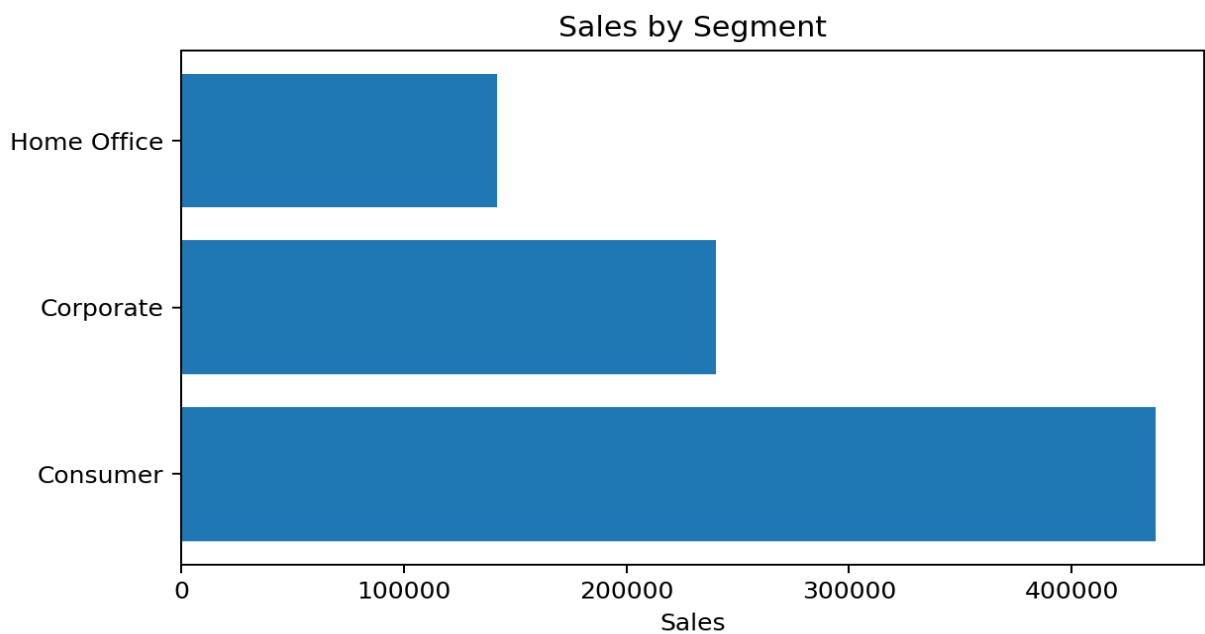
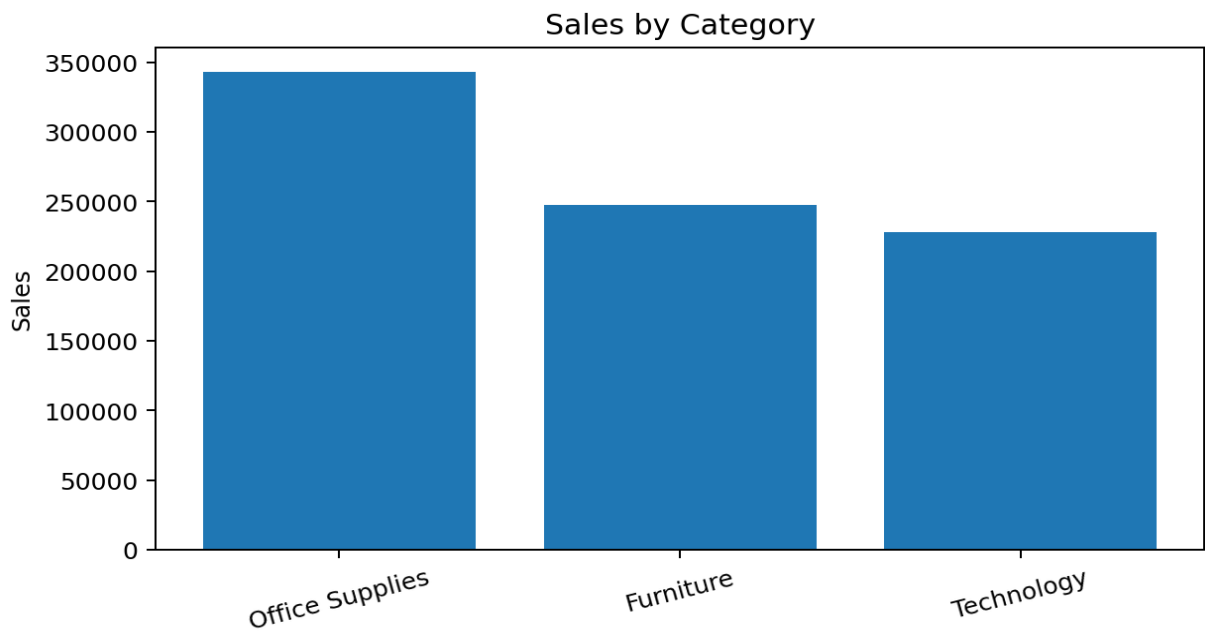


2. Sales Trend Analysis

Annual sales increased from **\$158,620** in 2014 to **\$272,538** in 2017. This indicates a strong long-term upward sales trend. The dashboard and automated pipeline can be used to monitor this trend continuously.

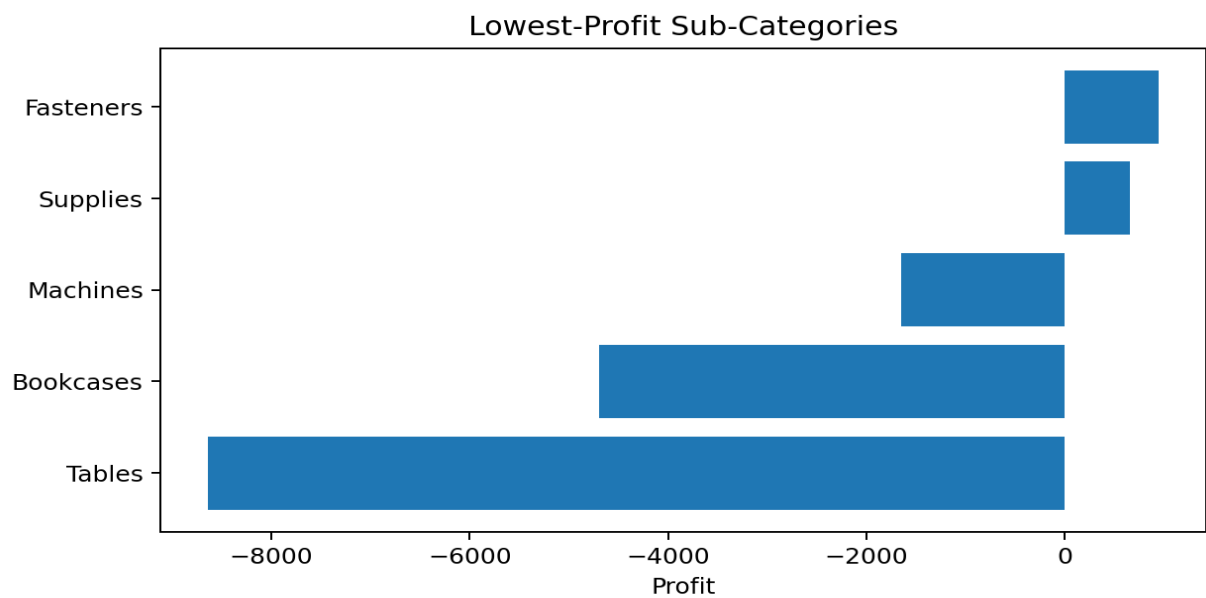
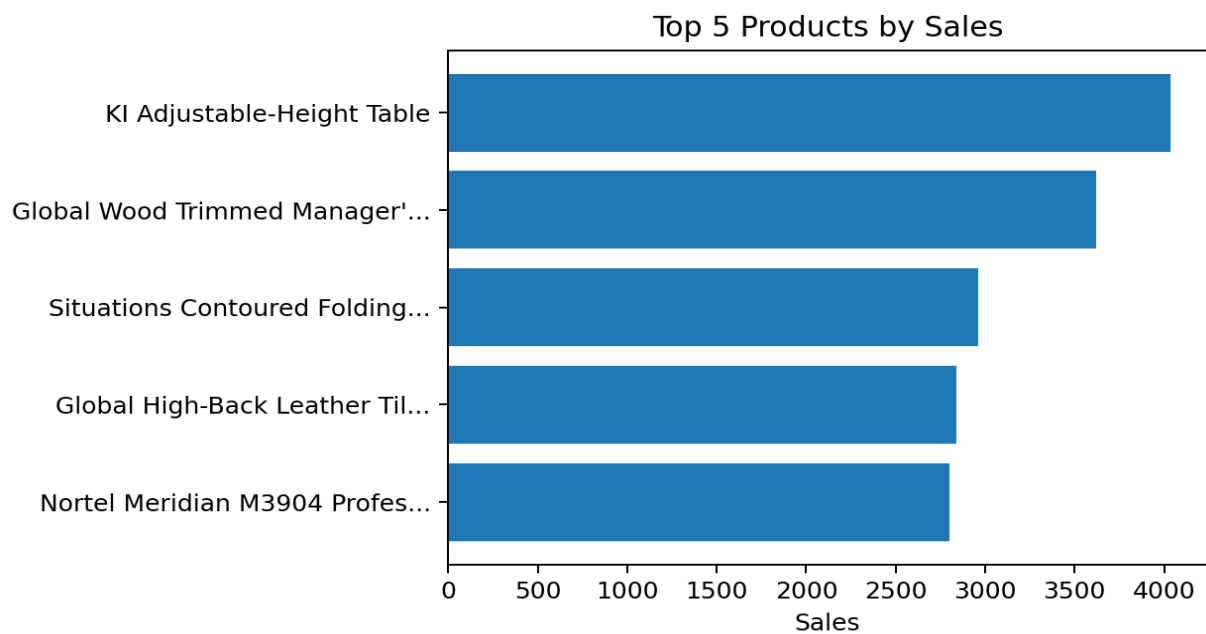


3. Category & Segment Performance



Insight: Office Supplies leads category sales at \$343,425. The Consumer segment is the largest segment with \$437,711 in sales. These areas should remain key targets for retention and growth initiatives.

4. Product & Profitability Insights



5. Top 5 Business Insights

1. West is the top region by sales with \$275,810; it should be studied for repeatable growth practices.
2. Consumer is the strongest customer segment with \$437,711 in sales and \$49,390 in profit.
3. Office Supplies is the largest category by sales at \$343,425.
4. Tables is the weakest sub-category by profit at \$-8,636; pricing, discounting and product costs should be reviewed.
5. Sales grew substantially over the four-year period, reaching \$272,538 in 2017, supporting continued investment in high-performing channels.

6. Three Business Recommendations

1. **Improve low-profit products:** Review Tables, Bookcases and Machines for discount levels, procurement costs and pricing. Reduce excessive discounting where it does not produce sufficient volume.
2. **Scale high-performing segments and regions:** Use the Consumer segment and West region as benchmarks for campaigns, retention programs and sales planning.
3. **Strengthen automated monitoring:** Run the pipeline on a schedule and refresh KPI/Excel outputs regularly so decision-makers can detect changes in sales, profit and product performance early.

7. Final Submission Checklist

Deliverable	Status
Final report PDF	Completed
Automation script	Completed
README documentation	Completed
requirements.txt	Completed
Processed Excel output	Completed
GitHub Actions workflow	Completed
PowerPoint presentation	Completed
LinkedIn final-submission post	Completed

Conclusion

The final Task 5 deliverable combines business analysis, dashboard-aligned insights and a repeatable automation pipeline. The project is ready for GitHub submission, final presentation and LinkedIn sharing.